AI-Guided Container Migration to Huawei Cloud CCE



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What AI Can and Cannot Do for Your Migration

Sharing of capabilities, not a formal delivery commitment

AI guidance for container migration to Huawei Cloud CCE

Central Note

AI accelerates container migration by automating discovery, chart conversion, and validation. It does not replace your judgment, your team's expertise, or customer approval. Use it as a powerful assistant — not as an autopilot.

1. Your Data Stays Yours

Security / Compliance / Protection

- AI does not need your real customer data
- Use simulated, test, or anonymized data
- Never upload credentials, keys, or sensitive data

2. AI Assists, You Decide

Reference / Human Review

- AI plans, analyzes, and writes charts
- Your team reviews and approves everything
- Test in staging before production migration

3. Get Customer Approval

Authorization / Approval

- The customer decides if AI is used in their project
- Without approval, do not enter customer details into AI
- Respect security, compliance, and confidentiality

4. When to Be Extra Careful

Prudent Evaluation / Manual Process

- Highly sensitive systems need extra review
- Financial core, defense, or classified workloads
- Consider manual migration for strict sovereignty rules

AI-assisted migration improves efficiency; it does not replace human review or customer authorization. Always validate in your specific environment before production cutover.

Why Container Migration Is Hard



What Makes It Difficult

Containers seem simple until you try to move them. Each cloud has its own Kubernetes service with different networking, storage, and ingress. Your Helm charts, image references, and config maps all need to change. Doing this by hand across dozens of services is slow and error-prone.



What Usually Goes Wrong

Services depend on each other in ways that are not obvious. Persistent volumes, secrets, and config maps get lost. Image references point to the old registry. Ingress rules do not match the new cluster. These issues surface only at cutover — the worst time to discover them.



How AI Changes This

AI reads your entire source cluster, understands the dependencies, and generates ready-to-deploy Helm charts for CCE. It converts images, updates references, and validates everything before you deploy. You describe what you want — AI handles the technical details.

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How Your AI Assistant Approaches the Migration

Step 1: Discover What You Have

Give AI access to your source Kubernetes cluster. It lists every deployment, service, config map, secret, persistent volume, and ingress rule — and maps the dependencies between them.

Example prompt: "Scan my Kubernetes cluster and list all resources with their dependencies."

Step 2: Convert for CCE

AI converts your manifests to CCE-compatible Helm charts, re-tags container images to Huawei Cloud SWR, and generates Terraform for the target cluster. You review everything before deploying.

Example prompt: "Convert these Kubernetes manifests to CCE Helm charts and push images to SWR."

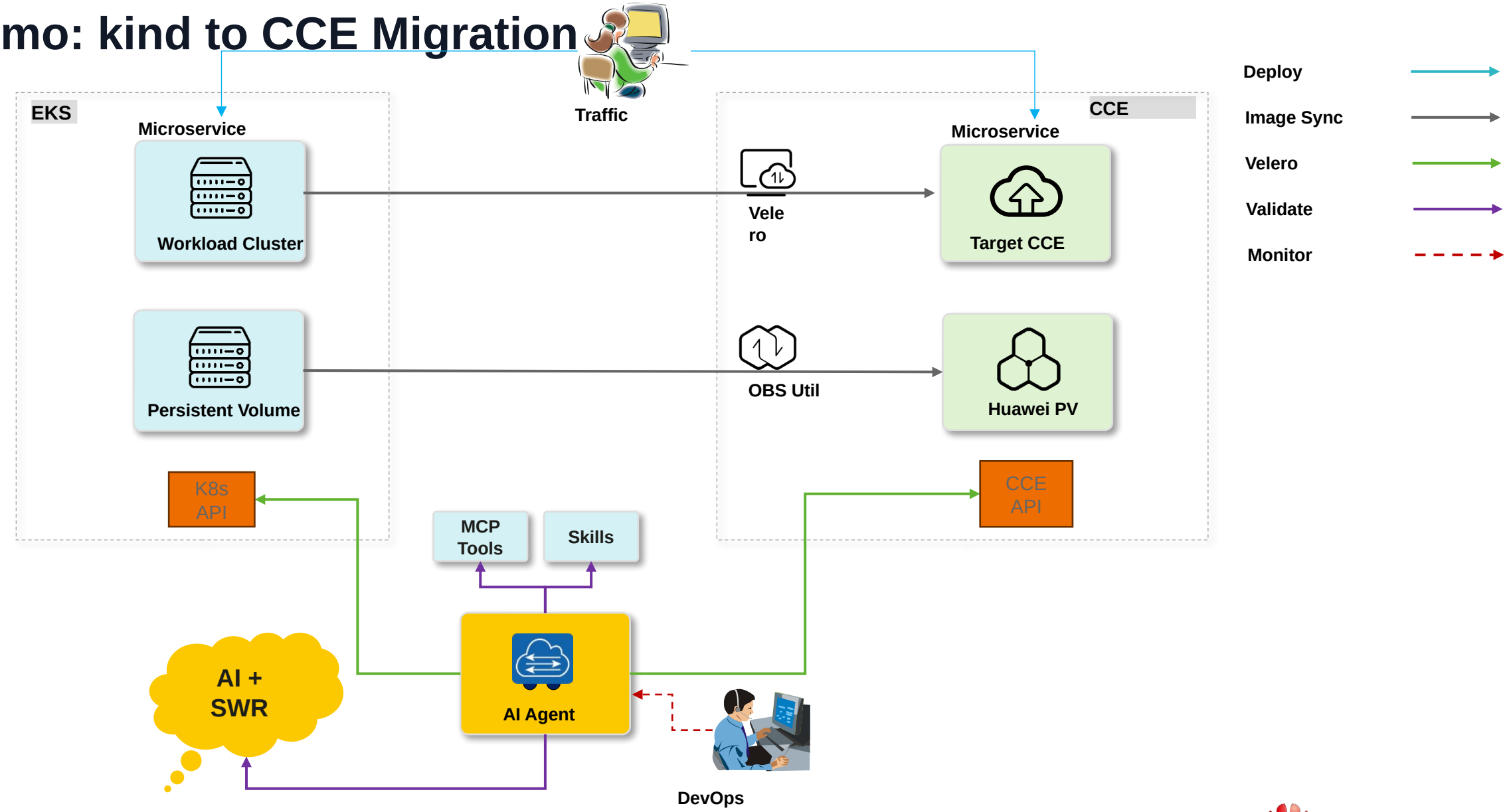
Step 3: Migrate and Validate

AI deploys to CCE, migrates persistent volumes via OBS, and verifies that every pod is running, every ingress is reachable, and every data path is intact. Your source cluster stays up until you confirm everything works.

Example prompt: "Deploy to CCE and validate all workloads are healthy."

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Demo: kind to CCE Migration



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Service Mapping: Source Cloud to Huawei Cloud

Source Cloud	Huawei Cloud	What Happens	Notes
Kubernetes (EKS/AKS/GKE)	CCE	Cluster config converted to CCE	K8s manifest conversion
Container Registry (ECR/ACR)	SWR	Images copied and re-tagged	Azure -> Huawei mapping
Persistent Volumes	EVS via OBS	Data backed up and restored	GCP -> Huawei mapping
Load Balancer	ELB	Ingress rules recreated	Container image migration
Monitoring	CES + AOM	Set up after migration	Image re-tagging
IAM Roles	IAM	Policies translated automatically	PV migration
VPC / Subnets	VPC / Subnets	Network recreated with Terraform	Load balancer config
Helm Charts	CCE Helm Charts	AI converts and validates	Monitoring setup
CI/CD Pipelines	CodeArts	Pipelines adapted for CCE	Role/policy translation
Auto-scaling	CCE Auto-scaler	Right-sized by AI	Network re-creation
—	CCI	Serverless option for batch workloads	App performance monitoring

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Your Migration Toolkit

SWR (Image Registry)

What it does:
- Hosts your container images
- Private and public repositories
- Scans images for vulnerabilities
- Triggers CCE deployments automatically

Ask AI about it:
"Which images need to go to SWR?"



Velero + OBS (Backup & Restore)

What it does:
- Backs up entire cluster to OBS
- Restores to a new CCE cluster
- Namespace-level granularity
- Works across regions

Ask AI about it:
"Back up my cluster with Velero"



CCI (Serverless Containers)

What it does:
- No node management needed
- Pay per second, scale to zero
- Perfect for batch and event-driven
- Integrates with CCE

Ask AI about it:
"Should I use CCI or CCE for this workload?"



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What You Gain with AI-Guided Migration

Save Time and Effort

AI converts Helm charts in hours, not days.
It right-sizes resources based on actual usage.
Your team focuses on decisions, not YAML files.
Natural language Kubernetes operations via AI.



Reduce Costs and Risk

AI recommends the most cost-effective CCE configuration.
Velero backup ensures zero data loss.
AI detects dependency issues before migration.
Keep your source cluster running until you confirm.



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Resources & Next Steps

Resources

GitHub:
binrogithub/1-3-Cloud-Adoption-Skills/AI-Assist-Migration/Container-Migration

Next Steps

1. Assess: inventory your workloads.
2. Test locally: use kind with AI.
3. Pilot: migrate one workload to CCE.
4. Production: deploy with Terraform.
5. Optimize: let AI tune autoscaling.

Contact

Huawei Cloud LATAM SMB team for CCE migration support

Container Migration with AI

AI reads your source cluster, converts to CCE Helm charts, pushes images to SWR, and validates everything.
Your source cluster stays running until you confirm the migration is complete.

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What You Gain with AI-Guided Migration



Save Time and Effort

AI converts Helm charts in hours, not days. It eliminates manual errors in manifest conversion. Your team focuses on architecture decisions, not YAML editing.



Reduce Migration Costs

AI recommends the most cost-effective CCE resources for your workloads. It identifies what can be optimized during migration.



Catch Problems Early

AI spots dependency issues, missing config maps, and broken image references before you deploy. No surprises at cutover.

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Thank you.

Bring digital to every person, home and organization for a fully connected, intelligent world.

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